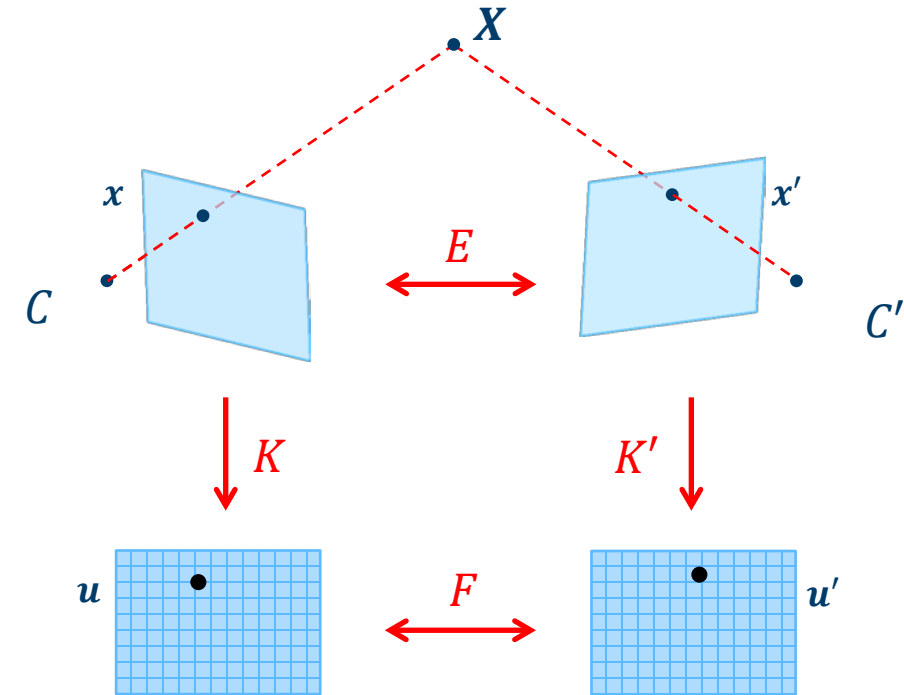


Lecture 7.3

Pose from epipolar geometry

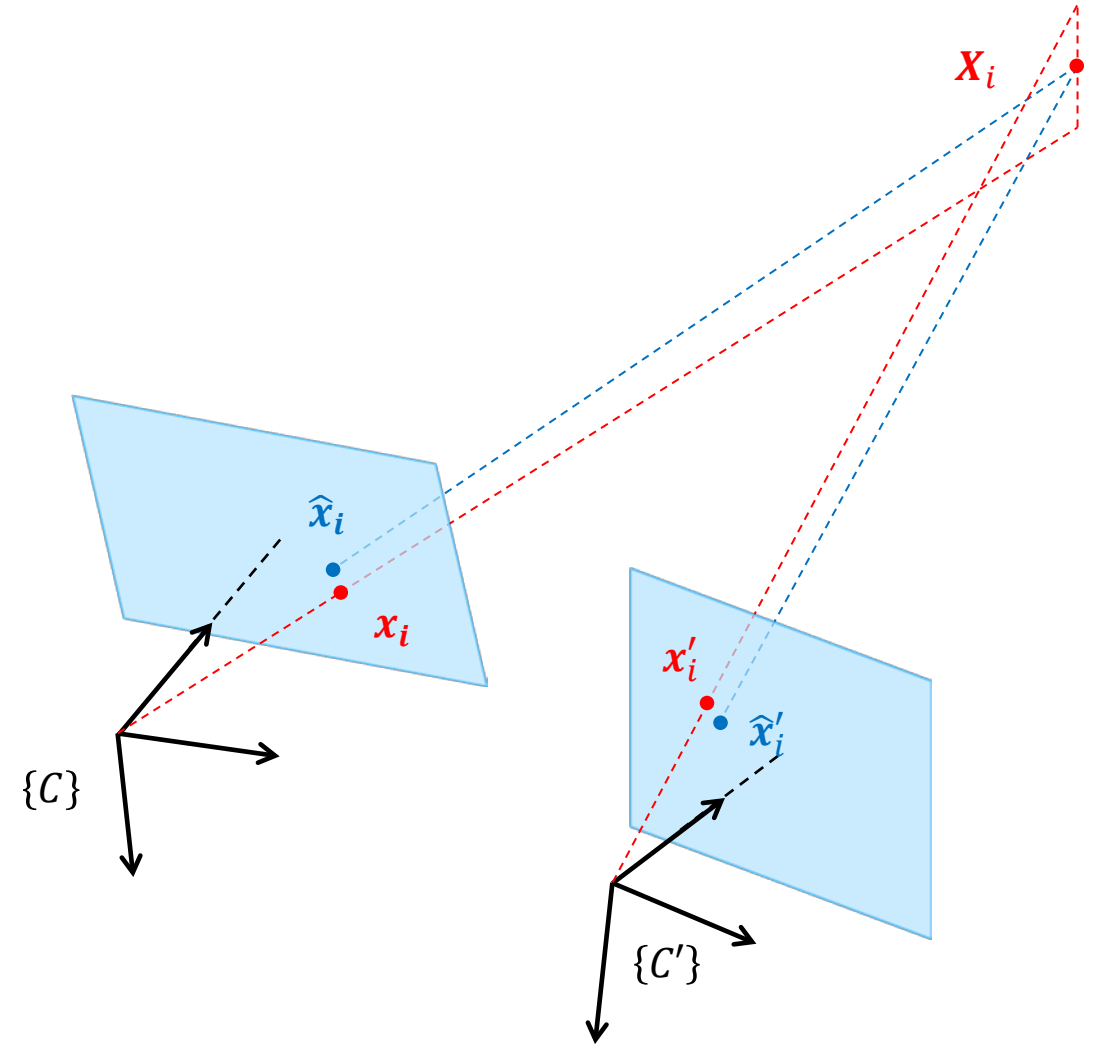
Introduction

- Representing epipolar geometry
 - The essential matrix $E = [t]_{\times} R$
 $\tilde{x}'^T E \tilde{x} = 0$
 - The fundamental matrix $F = K'^{-T} E K^{-1}$
 $\tilde{u}'^T F \tilde{u} = 0$
- Estimating epipolar geometry
 - F from 7 or 8 2D correspondences $u_i \leftrightarrow u_i'$
 - E from 5 2D correspondences $x_i \leftrightarrow x_i'$



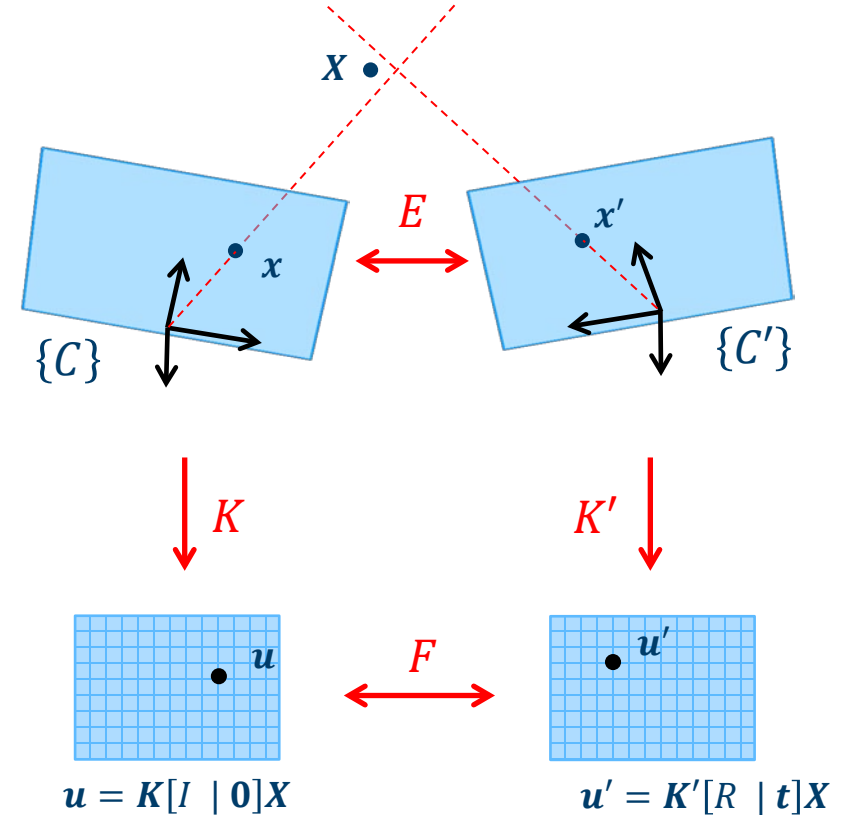
Introduction

- Representing epipolar geometry
 - The essential matrix $E = [t]_{\times} R$
 $\tilde{x}'^T E \tilde{x} = 0$
 - The fundamental matrix $F = K'^{-T} E K^{-1}$
 $\tilde{u}'^T F \tilde{u} = 0$
- Estimating epipolar geometry
 - F from 7 or 8 2D correspondences $u_i \leftrightarrow u_i'$
 - E from 5 2D correspondences $x_i \leftrightarrow x_i'$
- Exploiting epipolar geometry
 - 3D reconstruction of the scene by triangulation when camera matrices are known
- Now we will look at another way to make use of the epipolar geometry



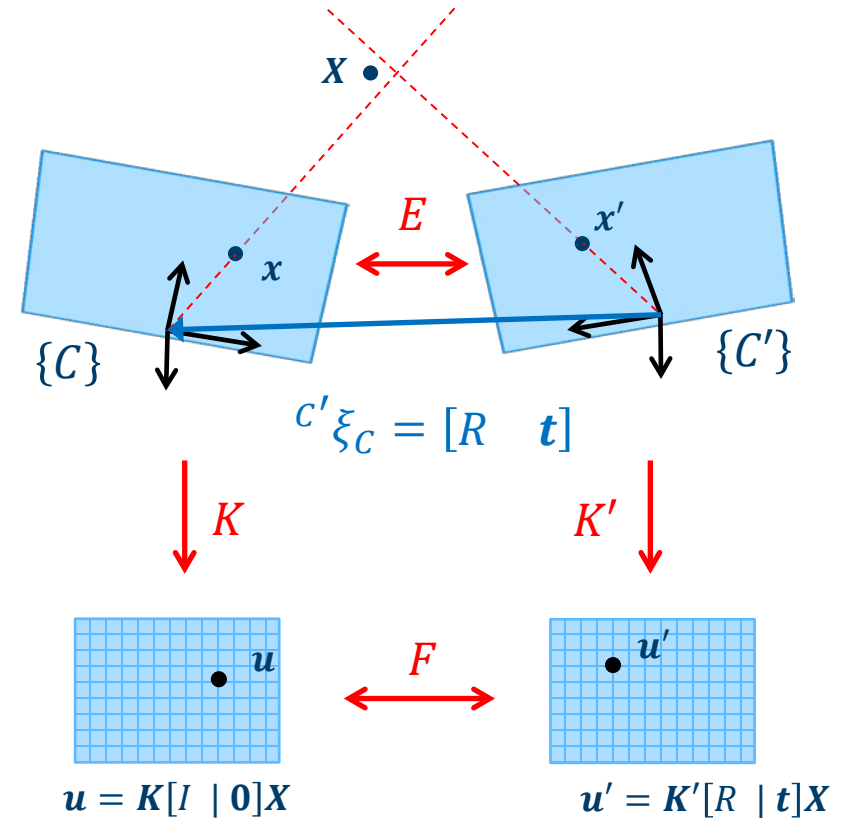
Pose from epipolar geometry

- If we restrict ourselves to calibrated cameras, the ambiguity gets restricted as well
- In the calibrated case, we can estimate and represent the epipolar geometry in terms of the essential matrix E
- By estimation, E will be a homogeneous matrix only restricted by
$$\tilde{x}'^T E \tilde{x} = 0$$
- But we also know that we can construct E non-homogeneously as $E = [t]_{\times} R$



Pose from epipolar geometry

- In 1981 H. C Longuet-Higgins* proved that one could recover the relative pose ${}^{C'}\xi_C = [R \ t]$ from the essential matrix up to the scale of t
- He argued that, up to the scale of t , there are 4 theoretical solutions, but only 1 for which the scene points would be in front of both cameras
 - This additional constraint has later been named the *cheirality constraint*



* H. C Longuet-Higgins, *A computer algorithm for reconstructing a scene from two projections*, 1981

Pose from epipolar geometry

- Since we only can estimate E up to scale, we can always rescale it so that the SVD of E has the form

$$E = UDV^T = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3] \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \mathbf{v}_3^T \end{bmatrix}$$

where $\det(U) = \det(V) = 1$

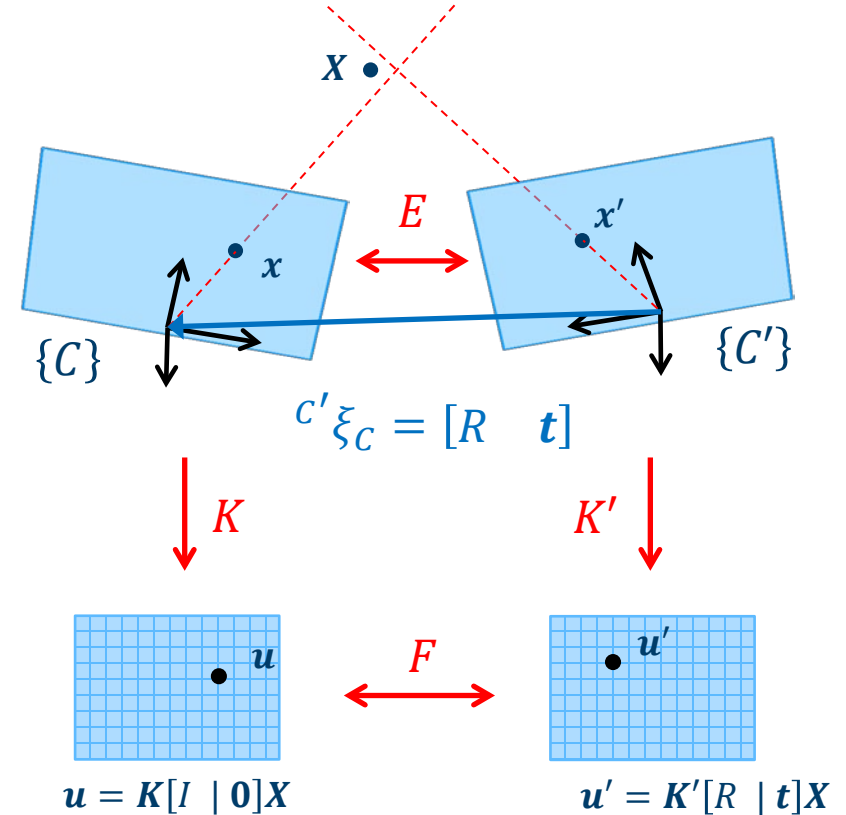
- Then one can show that

$$R \in \{UWV^T, UW^TV^T\}$$

$$\mathbf{t} = \pm\lambda\mathbf{u}_3; \lambda \in \mathbb{R} \setminus 0$$

where

$$W = \begin{bmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



Pose from epipolar geometry

- So the 4 candidate poses are

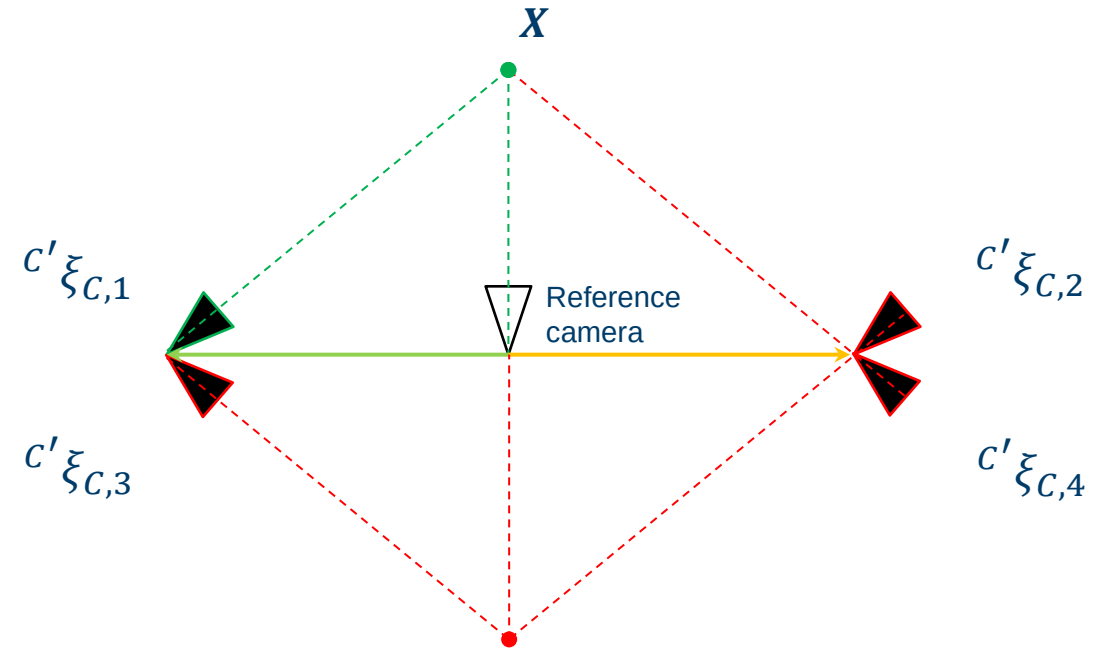
$$c'_{\xi_{C,1}} = [UWV^T \quad \mathbf{u}_3]$$

$$c'_{\xi_{C,2}} = [UWV^T \quad -\mathbf{u}_3]$$

$$c'_{\xi_{C,3}} = [UW^TV^T \quad \mathbf{u}_3]$$

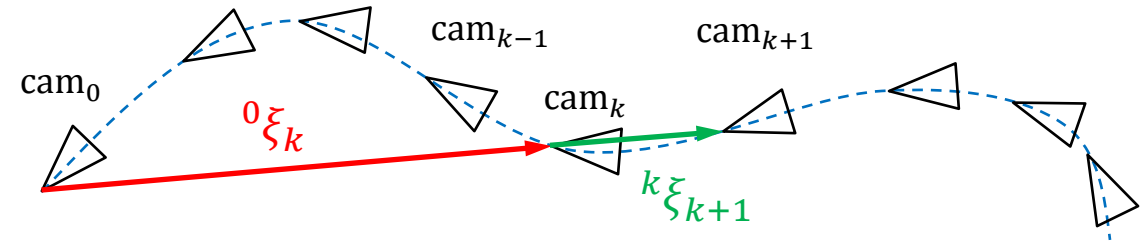
$$c'_{\xi_{C,4}} = [UW^TV^T \quad -\mathbf{u}_3]$$

- Their relation is shown in the figure for the case when $c'_{\xi_{C,1}}$ is the correct pose
- In general there is no way of knowing the correct candidate without imposing the cheirality constraint
- In theory it suffices to triangulate a single scene point X in order to determine the correct pose, but for robustness several points could be checked



Visual odometry

- Based on what we now know it is possible to do visual odometry, i.e. estimating the motion of a single camera from captured images
- The algorithm could look something like this



Visual odometry from 2D-correspondences

1. Capture new frame img_{k+1}
2. Extract and match features between img_{k+1} and img_k
3. Estimate the essential matrix $E_{k,k+1}$
4. Decompose the $E_{k,k+1}$ into ${}^kR_{k+1}$ and ${}^kt_{k+1}$ to get the relative pose

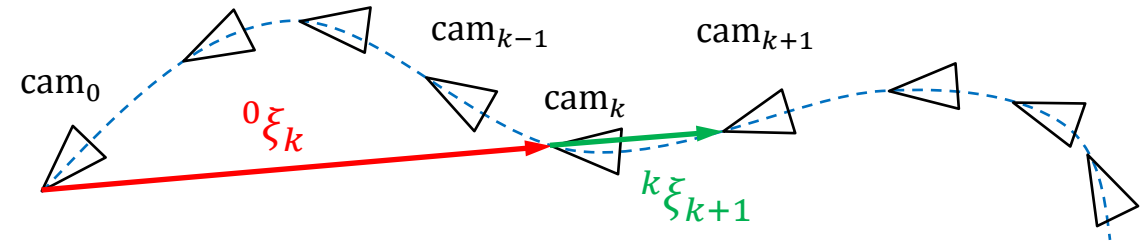
$${}^k\xi_{k+1} = [{}^kR_{k+1} \quad {}^kt_{k+1}]$$

5. Calculate the pose of camera $k + 1$ relative to the first camera

$${}^0\xi_{k+1} = {}^0\xi_k {}^k\xi_{k+1}$$

Visual odometry

- Based on what we now know it is possible to do visual odometry, i.e. estimating the motion of a single camera from captured images
- The algorithm could look something like this
 - Neglects the unknown scale of ${}^k\mathbf{t}_{k-1}$
 - We should set $\|{}^1\mathbf{t}_0\| = 1$ and scale the other translations accordingly



Visual odometry from 2D-correspondences

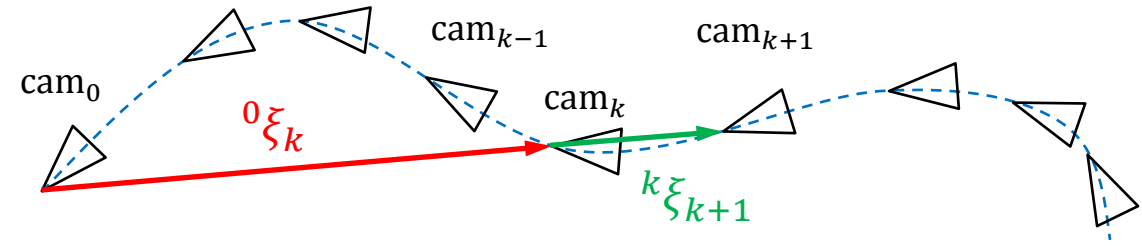
- Capture new frame img_{k+1}
- Extract and match features between img_{k+1} and img_k
- Estimate the essential matrix $E_{k,k+1}$
- Decompose the $E_{k,k+1}$ into ${}^kR_{k+1}$ and ${}^k\mathbf{t}_{k+1}$ to get the relative pose
- Calculate the pose of camera $k + 1$ relative to the first camera

$${}^k\xi_{k+1} = [{}^kR_{k+1} \quad {}^k\mathbf{t}_{k+1}]$$

$${}^0\xi_{k+1} = {}^0\xi_k {}^k\xi_{k+1}$$

Visual odometry

- Based on what we now know it is possible to do visual odometry, i.e. estimating the motion of a single camera from captured images
- A better visual odometry algorithm can look like this



Visual odometry from 2D-correspondences

1. Capture new frame img_{k+1}
2. Extract and match features between img_{k+1} and img_k
3. Estimate the essential matrix $E_{k,k+1}$
4. Decompose the $E_{k,k+1}$ into ${}^kR_{k+1}$ and ${}^kt_{k+1}$ to get the relative pose

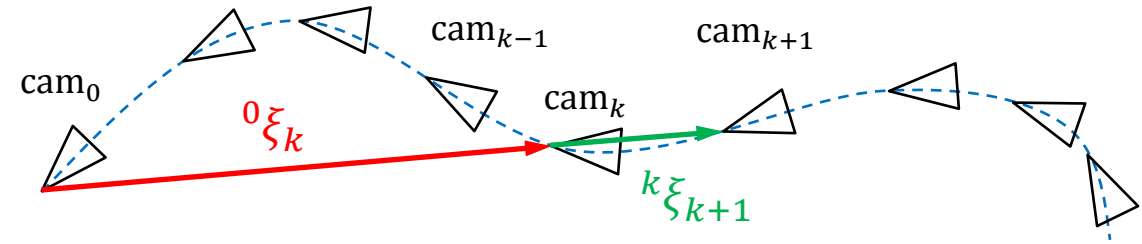
$${}^k\xi_{k+1} = [{}^kR_{k+1} \quad {}^kt_{k+1}]$$

5. Compute $\|{}^kt_{k+1}\|$ from $\|{}^{k-1}t_k\|$ and rescale ${}^kt_{k+1}$ accordingly
6. Calculate the pose of camera $k + 1$ relative to the first camera

$${}^0\xi_{k+1} = {}^0\xi_k {}^k\xi_{k+1}$$

Visual odometry

- Based on what we now know it is possible to do visual odometry, i.e. estimating the motion of a single camera from captured images
- A better visual odometry algorithm can look like this
 - How to compute $\|{}^{k+1}\mathbf{t}_k\|$ from $\|{}^k\mathbf{t}_{k-1}\|$?



Visual odometry from 2D-correspondences

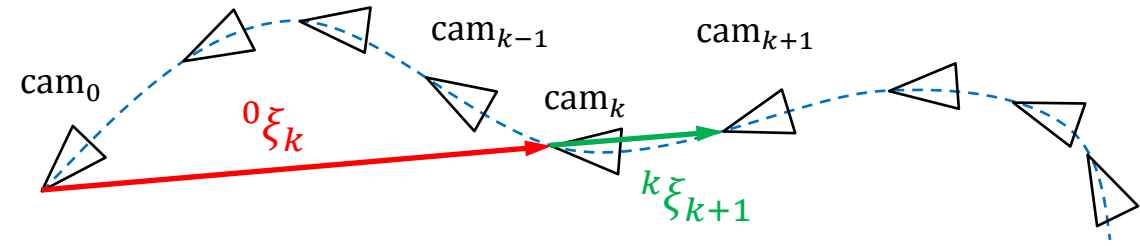
1. Capture new frame img_{k+1}
2. Extract and match features between img_{k+1} and img_k
3. Estimate the essential matrix $E_{k,k+1}$
4. Decompose the $E_{k,k+1}$ into ${}^kR_{k+1}$ and ${}^k\mathbf{t}_{k+1}$ to get the relative pose
5. Compute $\|{}^k\mathbf{t}_{k+1}\|$ from $\|{}^{k-1}\mathbf{t}_k\|$ and rescale ${}^k\mathbf{t}_{k+1}$ accordingly
6. Calculate the pose of camera $k + 1$ relative to the first camera

$${}^0\xi_{k+1} = {}^0\xi_k {}^k\xi_{k+1}$$

Visual odometry

- Based on what we now know it is possible to do visual odometry, i.e. estimating the motion of a single camera from captured images
- A better visual odometry algorithm can look like this
 - How to compute $\|{}^{k+1}\mathbf{t}_k\|$ from $\|{}^k\mathbf{t}_{k-1}\|$?
 - Determine two scene points ${}^k\mathbf{X}_{k-1,k}$ and ${}^k\mathbf{X}'_{k-1,k}$ by triangulation of two 2D-correspondences ${}^{k-1}\mathbf{x} \leftrightarrow {}^k\mathbf{x}$ and ${}^{k-1}\mathbf{x}' \leftrightarrow {}^k\mathbf{x}'$
 - Determine the same two scene points ${}^k\mathbf{X}_{k,k+1}$ and ${}^k\mathbf{X}'_{k,k+1}$ by triangulation of two 2D-correspondences ${}^k\mathbf{x} \leftrightarrow {}^{k+1}\mathbf{x}$ and ${}^k\mathbf{x}' \leftrightarrow {}^{k+1}\mathbf{x}'$
 - Then

$$\frac{\|{}^{k-1}\mathbf{t}_k\|}{\|{}^k\mathbf{t}_{k+1}\|} = \frac{\|{}^k\mathbf{X}_{k-1,k} - {}^k\mathbf{X}'_{k-1,k}\|}{\|{}^k\mathbf{X}_{k,k+1} - {}^k\mathbf{X}'_{k,k+1}\|}$$



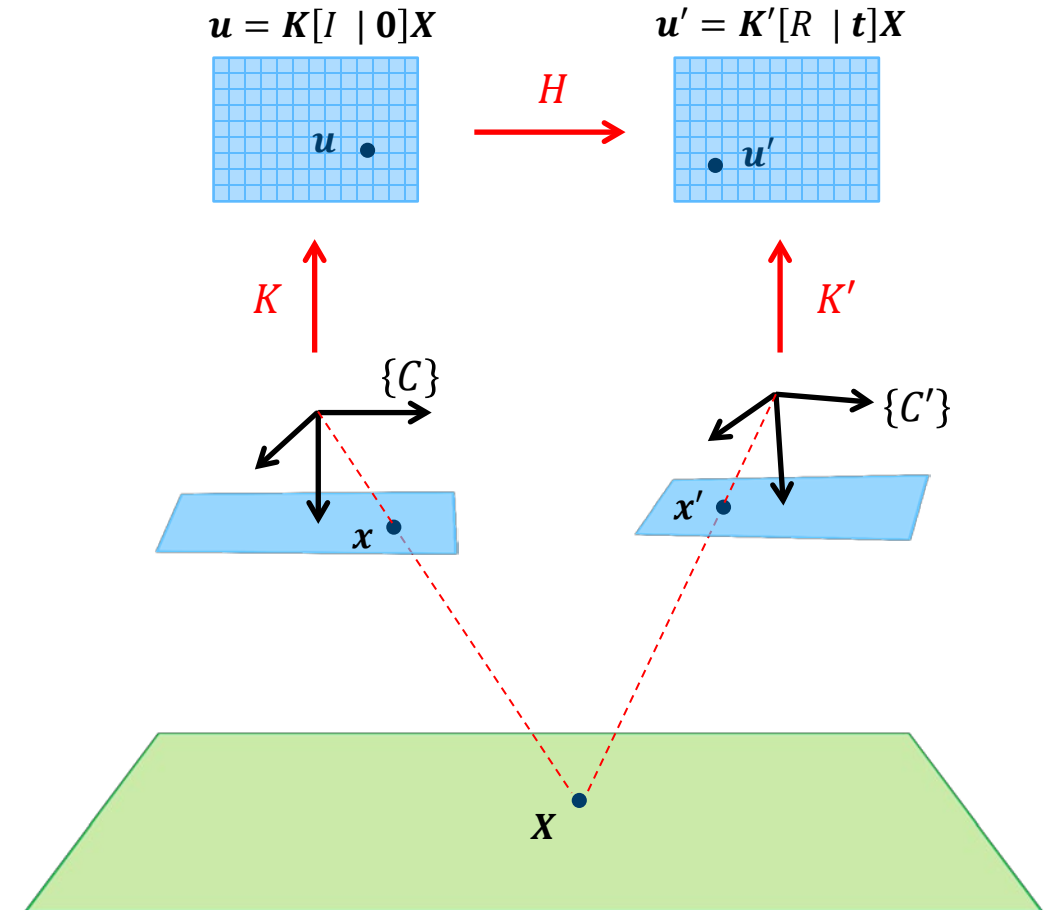
Visual odometry from 2D-correspondences

- Capture new frame img_{k+1}
- Extract and match features between img_{k+1} and img_k
- Estimate the essential matrix $E_{k,k+1}$
- Decompose the $E_{k,k+1}$ into ${}^kR_{k+1}$ and ${}^k\mathbf{t}_{k+1}$ to get the relative pose
- Compute $\|{}^k\mathbf{t}_{k+1}\|$ from $\|{}^{k-1}\mathbf{t}_k\|$ and rescale ${}^k\mathbf{t}_{k+1}$ accordingly
- Calculate the pose of camera $k + 1$ relative to the first camera

$${}^0\xi_{k+1} = {}^0\xi_k {}^k\xi_{k+1}$$

Planar scene

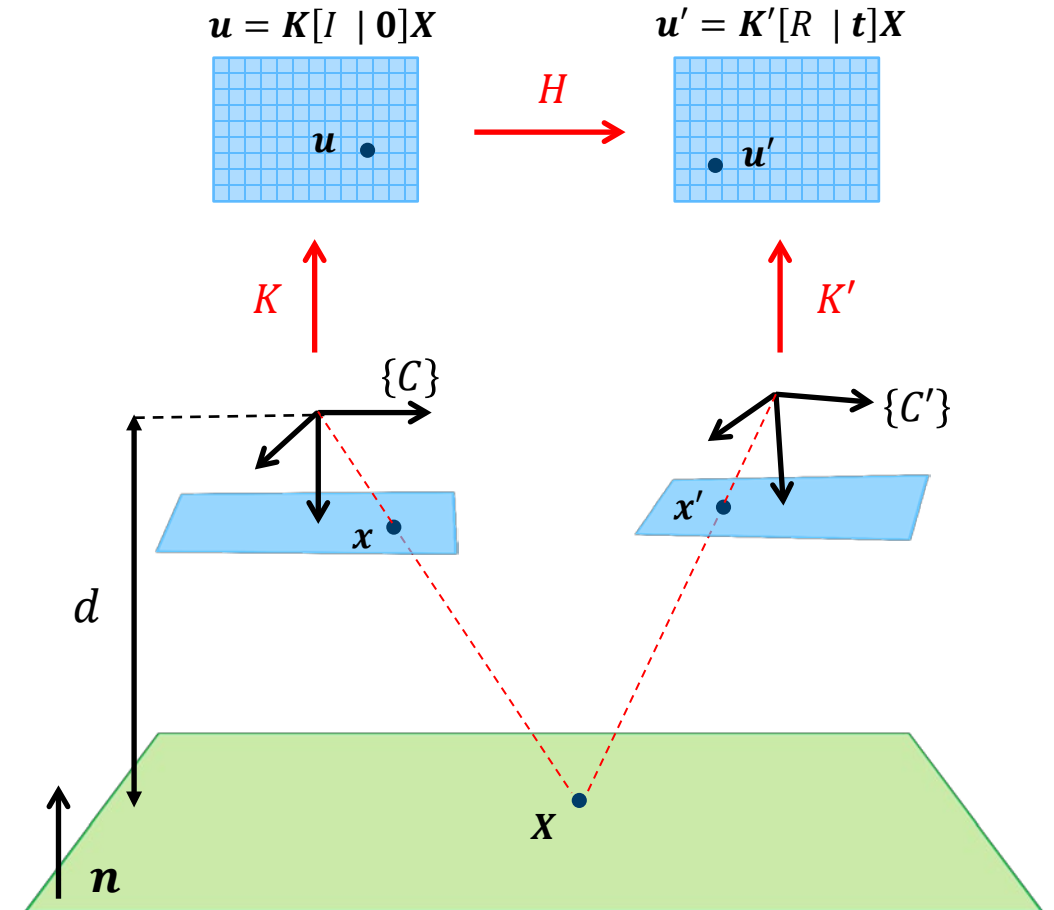
- When the scene is planar, it is not possible to estimate the epipolar geometry between two views from 2D-correspondences
 - In the case of an almost planar scene, the estimation is likely to be **ill-conditioned**
- We know that for this case the relationship between image points and scene points can be described by homographies



Planar scene

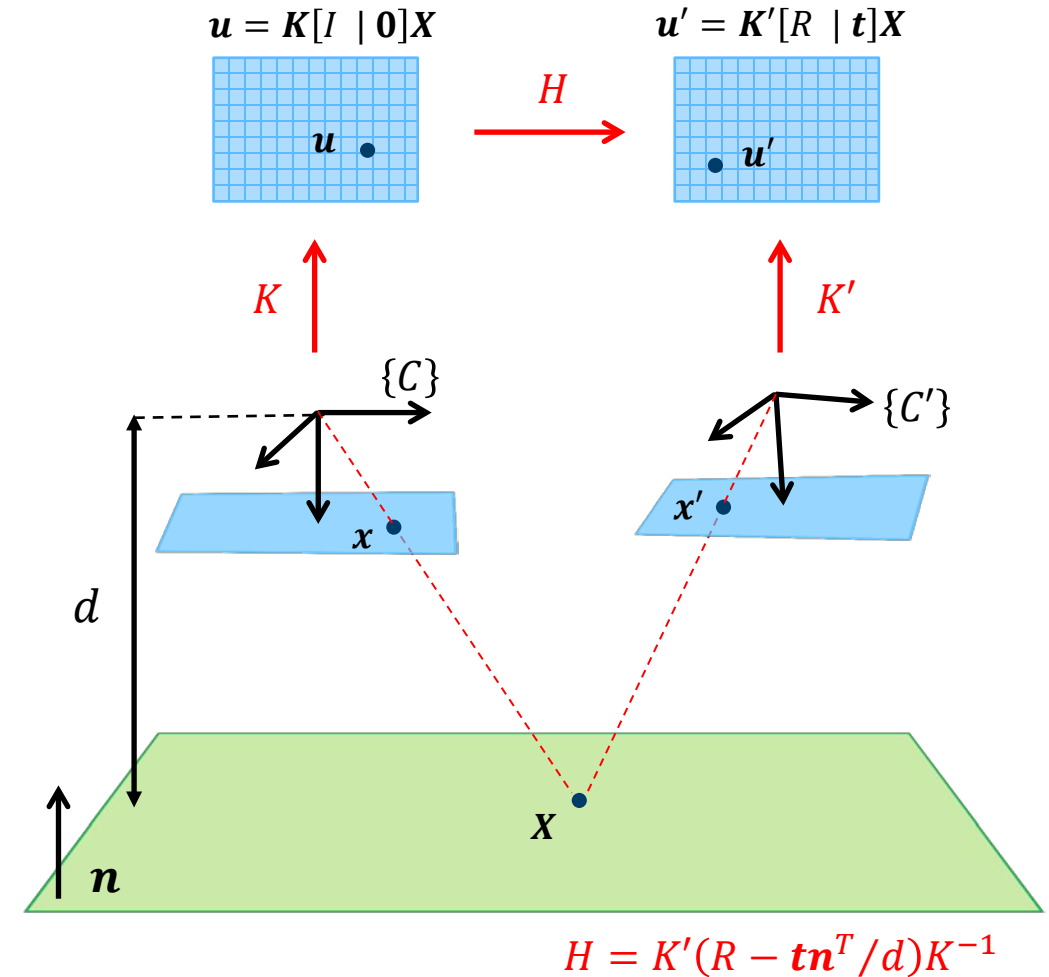
- When the scene is planar, it is not possible to estimate the epipolar geometry between two views from 2D-correspondences
 - In the case of an almost planar scene, the estimation is likely to be **ill-conditioned**
- We know that for this case the relationship between image points and scene points can be described by homographies
- If the relative pose between views are ${}^{C'}\xi_C = [R \quad \mathbf{t}]$, then one can show that the homography between views must be given by
$$H = K'(R - \mathbf{t}\mathbf{n}^T/d)K^{-1}$$

where $\mathbf{n}$ is the normal vector of the plane and d is the depth of the plane relative to $\{C\}$



Planar scene

- Based on the expression $H = K'(R - \mathbf{t}\mathbf{n}^T/d)K^{-1}$ it is possible to **estimate $(R, \mathbf{n}, \mathbf{t}/d)$ from a known homography** in a process known as homography decomposition
 - So for situations where we know the plane depth d , we get the relative pose ${}^{C'}\xi_C = [R \quad \mathbf{t}]$
- In the 2007 report *Deeper understanding of the homography decomposition for vision-based control*, Ezio Malis & Manuel Vargas derive that the decomposition problem has 4 analytical solutions
 - Two solutions can be invalidated by requiring points to be in front of the cameras
 - With some knowledge about the normal vector $\mathbf{n}$ it is often possible to eliminate one out of the two remaining solutions
 - OpenCV – `cv::decomposeHomographyMat`



Planar scene

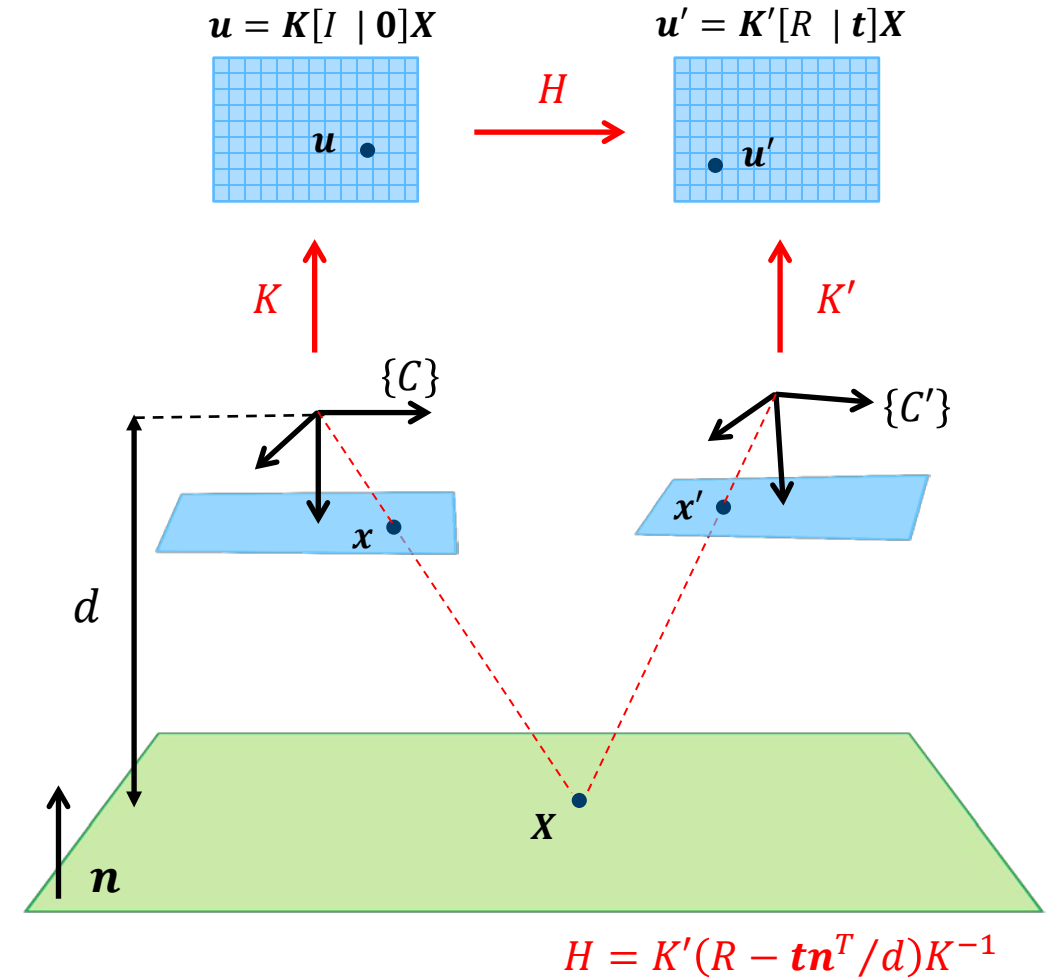
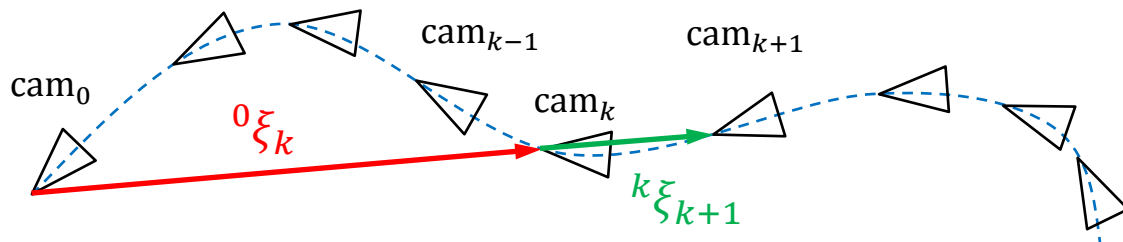
Visual odometry from 2D-correspondences , planar case with known plane depth

1. Capture new frame img_{k+1}
2. Extract and match features between img_{k+1} and img_k
3. Estimate homography $H_{k,k+1}$
4. Decompose the $H_{k,k+1}$ and eliminate 3 out of the 4 possible solutions to get the relative pose

$${}^k\xi_{k+1} = [{}^kR_{k+1} \quad {}^k\mathbf{t}_{k+1}]$$

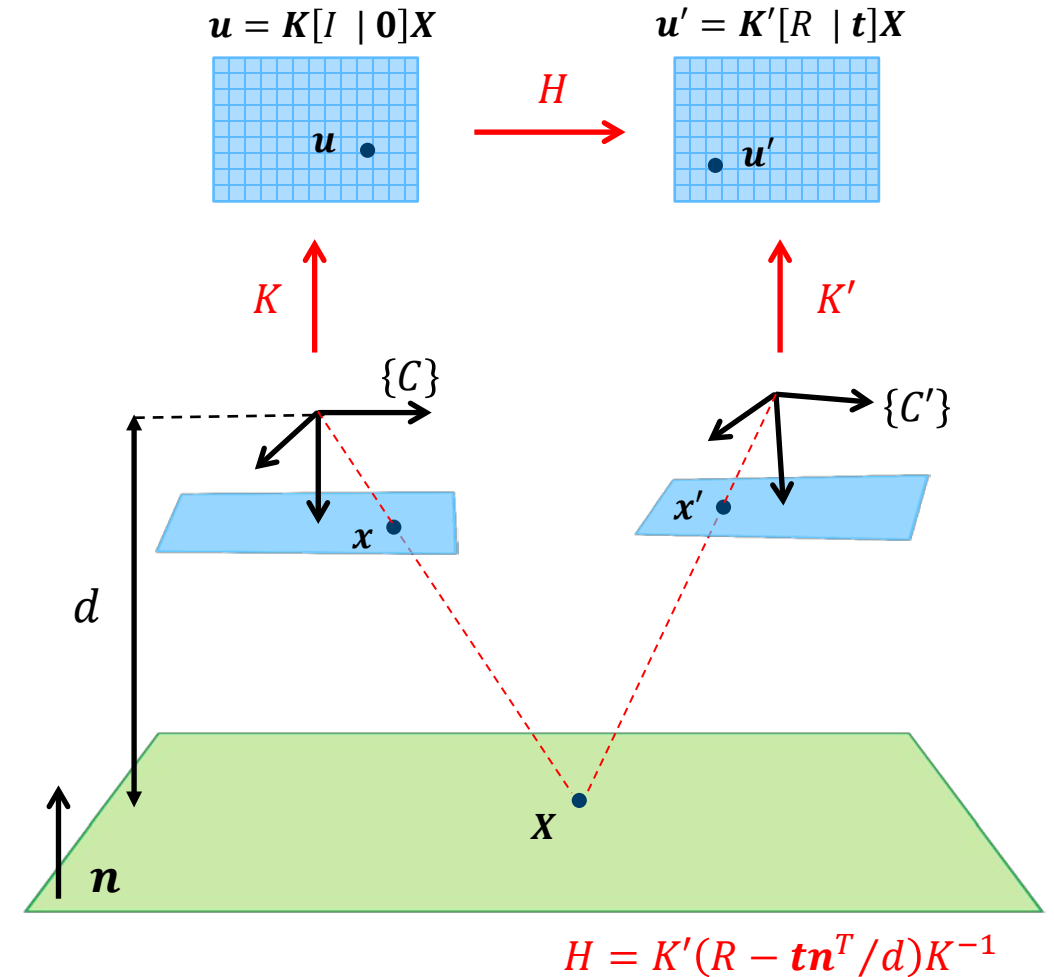
5. Calculate the pose of camera $k + 1$ relative to the first camera

$${}^0\xi_{k+1} = {}^0\xi_k {}^k\xi_{k+1}$$



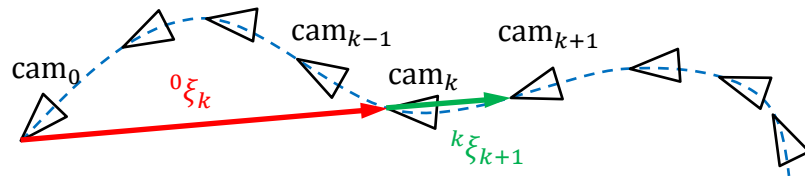
Planar scene

- This method is well suited for environments where we can detect scene planes and have some knowledge about the orientation of these planes
- Indoors
 - walls, floor, ceiling
- In city environments
 - ground, sides of buildings
- Imaging from high altitudes
 - ground



Summary

- Pose from epipolar geometry
 - Non-planar case
 - Estimate epipolar geometry
 - Estimate relative pose from E
 - Planar case
 - Estimate homography
 - Estimate relative pose from H
 - Visual odometry
- Additional reading:
 - Szeliski: 7.2
 - Optional reading:
 - H. C Longuet-Higgins, *A computer algorithm for reconstructing a scene from two projections*, 1981
 - Ezio Malis & Manuel Vargas, *Deeper understanding of the homography decomposition for vision-based control*, 2007
 - Davide Scaramuzza, *Tutorial on Visual Odometry*

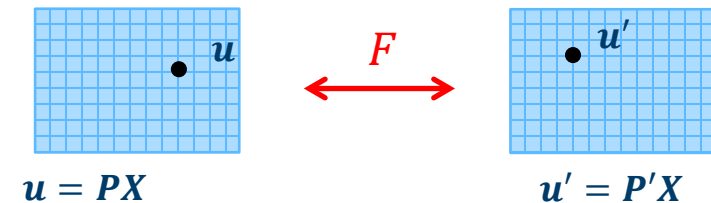
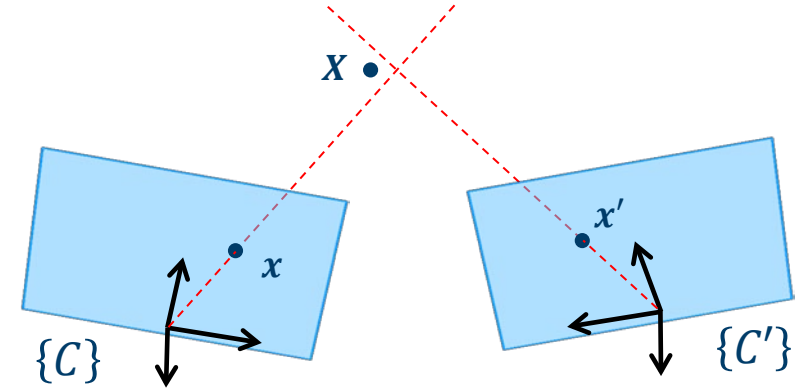


Recall the transformations of projective space

Transformation of $\mathbb{P}^3$	Matrix	#DoF	Preserves
Translation	$\begin{bmatrix} I & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	3	Orientation + all below
Euclidean	$\begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	6	Volumes, volume ratios, lengths + all below
Similarity	$\begin{bmatrix} sR & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$	7	Angles + all below
Affine	$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ 0 & 0 & 0 & 1 \end{bmatrix}$	12	Parallelism of planes, The plane at infinity + all below
Homography /projective	$\begin{bmatrix} h_{11} & h_{12} & h_{13} & h_{14} \\ h_{21} & h_{22} & h_{23} & h_{24} \\ h_{31} & h_{32} & h_{33} & h_{34} \\ h_{41} & h_{42} & h_{43} & h_{44} \end{bmatrix}$	15	Intersection and tangency of surfaces in contact, straight lines

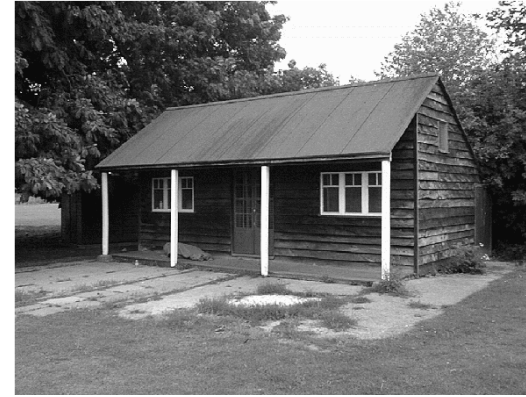
Pose from epipolar geometry

- One of the most important results in computer vision is that it is possible to determine the camera matrices P and P' that correspond to a given fundamental matrix F
 - Not uniquely, but up to a projective ambiguity, so lengths, angles or parallel lines/surfaces are not preserved
- If H is a projective transformation of 3-space, then the fundamental matrix corresponding to the pair of camera matrices (P, P') is the same as to that of (HP, HP')
- This can still be used to estimate the structure of the scene by triangulation, but the reconstructed scene would also suffer from a projective ambiguity

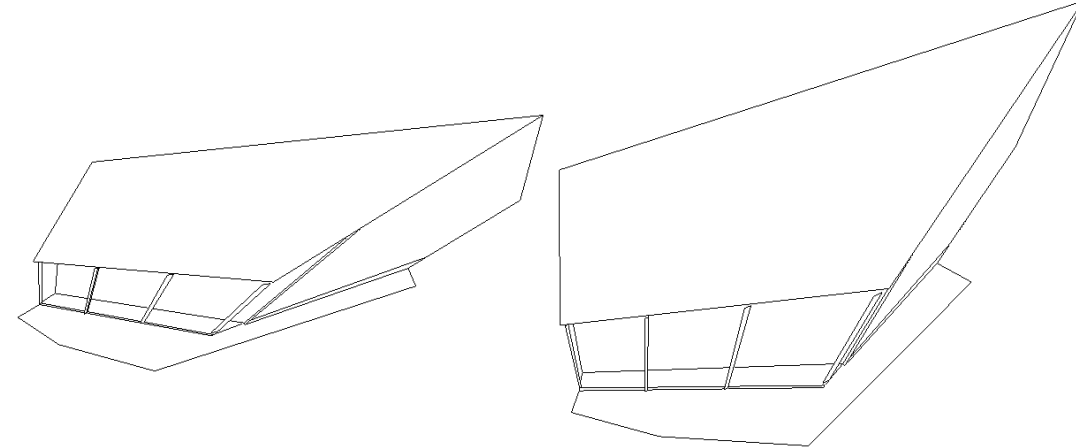


Pose from epipolar geometry

- One of the most important results in computer vision is that it is possible to determine the camera matrices P and P' that correspond to a given fundamental matrix F
 - Not uniquely, but up to a projective ambiguity, so lengths, angles or parallel lines/surfaces are not preserved
- If H is a projective transformation of 3-space, then the fundamental matrix corresponding to the pair of camera matrices (P, P') is the same as to that of (HP, HP')
- This can still be used to estimate the structure of the scene by triangulation, but the reconstructed scene would also suffer from a projective ambiguity

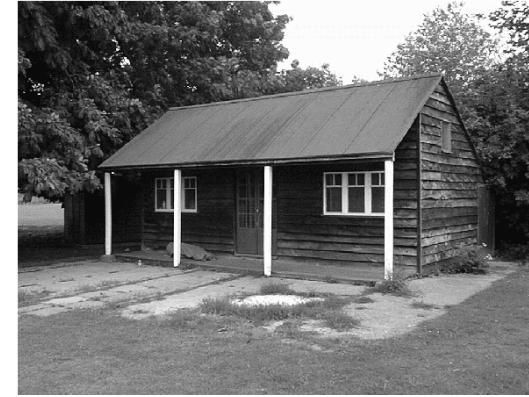


Images courtesy of Hartley & Zisserman <http://www.robots.ox.ac.uk/~vgg/hzbook/>



Pose from epipolar geometry

- One of the most important results in computer vision is that it is possible to determine the camera matrices P and P' that correspond to a given fundamental matrix F
 - Not uniquely, but up to a projective ambiguity, so lengths, angles or parallel lines/surfaces are not preserved
- If H is a projective transformation of 3-space, then the fundamental matrix corresponding to the pair of camera matrices (P, P') is the same as to that of (HP, HP')
- This can still be used to estimate the structure of the scene by triangulation, but the reconstructed scene would also suffer from a projective ambiguity
 - By adding knowledge about the scene the ambiguity can be restricted to affine or even metric



Images courtesy of Hartley & Zisserman <http://www.robots.ox.ac.uk/~vgg/hzbook/>

